ADVANCED NETWORK THREAT HUNTING AND INCIDENT INVESTIGATION

CYBERSECURITY PROJECT REPORT

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Investigation performed on a provided packet capture using Wireshark and supporting command-line tools.

1. Executive Summary

This investigation reviewed the supplied network capture to identify notable DNS, HTTP, TLS and TCP activity. The main confirmed finding was an HTTP download of runner_ilove.exe from 107.175.82.242 on TCP port 9000. The response returned HTTP 200 OK and a file size of 1,646,592 bytes. The recovered file matched the local copy by SHA-256. The executable was not run. The evidence confirms the download, but does not establish that the file is malicious.

2. Objectives

• Review DNS, HTTP, TLS and TCP traffic.

• Investigate the observed executable download.

• Record indicators and arrange observed events in a timeline.

• Prepare defensive display filters and example detection rules.

• Document limitations and practical security recommendations.

3. Scope and Method

The analysis was performed on the supplied capture file 2026-10-01-traffic-analysis-exercise.pcap. Tools included Wireshark, TShark and command-line file inspection utilities. The work was limited to the supplied traffic and recovered artifacts. No claim is made that every host or event in the environment was investigated.

The executable was inspected statically and was not executed.

4. Traffic Overview

Protocol / Metric	Observed count
Total frames	42,497
TCP	41,600
UDP	736
DNS	319
TLS	10,632
HTTP	32
ICMP	8
ARP	153

Main client observed: 10.10.1.128. Internal DNS/server address observed: 10.10.1.10.

5. Key Findings

5.1 Confirmed executable download

At 189.451928 seconds, frame 23091 recorded an HTTP GET request to 107.175.82.242:9000 for /wilow/runner_ilove.exe. The User-Agent was curl/8.21.0. At 191.225158 seconds, frame 24916 recorded HTTP 200 OK with Content-Length 1,646,592 and content type binary/octet-stream. The traffic was associated with TCP stream 128.

The recovered runner_from_pcap.exe matched runner_ilove.exe by size and SHA-256. This verifies the file match, not its maliciousness.

5.2 holtandberyl.com requests

The client queried holtandberyl.com at 92.641104 seconds. DNS resolved to 35.170.55.9. HTTP requests to /prw and /crv received 301 redirects to HTTPS, followed by TLS ClientHello messages using that hostname as SNI. The encrypted HTTPS content was not established from this evidence. These observations alone do not prove malicious activity.

5.3 Unresolved encrypted connection

Traffic to 195.63.128.106:443 began at 196.974101 seconds. Its purpose and hostname were not established. Its timing does not prove a relationship to the executable download.

5.4 DNS negative responses and cloud traffic

Queries for wpad.natureforce.com received NXDOMAIN responses. This can occur during normal proxy discovery and is an investigation lead rather than a verdict. Traffic involving vsblobprodscussu5shard51.blob.core.windows.net was also observed; cloud storage traffic may be legitimate and requires environmental context.

6. Static Analysis

The recovered file was identified as a PE32+ Windows GUI executable for x86-64 with eight sections and a size of 1,646,592 bytes. Static inspection found imports including CreateThread, VirtualAlloc, VirtualProtect, VirtualQuery, Sleep and WaitForSingleObject. Such functions can be used by legitimate or malicious software and do not independently establish malicious behavior.

SHA-256: b789adce70c028a6d17be45ec34667aa95586350ee0c4057e75b138b1ad35061

7. Event Timeline

Capture time (s)	Observed event
92.641104	DNS query for holtandberyl.com.
92.815468	DNS response included 35.170.55.9.
92.949426	HTTP request to /prw.
93.024330	HTTP 301 redirect to HTTPS /prw.
93.098277	TLS ClientHello with SNI holtandberyl.com.
94.383042	HTTP request to /crv.
94.453762	HTTP 301 redirect to HTTPS /crv.
94.537468	TLS ClientHello with SNI holtandberyl.com.
189.451928	GET request for runner_ilove.exe.

Capture time (s)	Observed event
191.225158	HTTP 200 OK; 1,646,592-byte response.
After recovery	SHA-256 comparison matched recovered and local files.

The relationship between holtandberyl.com traffic and the executable download has not been established.

8. Indicators of Interest

Type	Indicator	Status / interpretation
IP and Port	107.175.82.242:9000	Confirmed observation: Served runner_ilove.exe; maliciousness not confirmed
File	runner_ilove.exe	Confirmed observation: PE32+ executable; do not execute
SHA256	b789adce70c028a6d17be45ec34667aa95586350ee0c4057e75b138b1ad35061	Confirmed file hash: Matches the recovered and local file; not a malware verdict
Domain	holtandberyl.com	Unverified lead: HTTP requests to /prw and /crv redirected to HTTPS
IP	35.170.55.9	Unverified lead: Resolved for holtandberyl.com; purpose not established
Domain	vsblobprodscussu5shard51.blob.core.windows.net	Contextual observation: TLS SNI observed; could be legitimate cloud storage
IP	195.63.128.106:443	Unresolved lead: Encrypted data observed; hostname and purpose unknown
Domain	wpad.natureforce.com	Likely benign or inconclusive: NXDOMAIN responses; may relate to normal proxy discovery

9. Detection Filters

Executable server traffic: `ip.addr == 107.175.82.242 && tcp.port == 9000`

Observed HTTP paths: `http.host == "holtandberyl.com" && (http.request.uri == "/prw" || http.request.uri == "/crv")`

DNS error responses: `dns.flags.response == 1 && dns.flags.rcode != 0`

These filters support investigation. They should not be treated as proof of compromise without additional context.

10. Defensive Recommendations

• Review proxy, endpoint and firewall logs for the executable download and identify the initiating process and user.

• Check the file hash using approved endpoint security and malware-analysis tools without executing the file on a normal workstation.

• Review destination reputation and related DNS, HTTP and TLS events with trusted threat-intelligence sources.

• Investigate the connection to 195.63.128.106:443 using endpoint and network logs.

• Tune DNS and executable-download alerts to reduce false positives; validate example rules before deployment.

• Preserve the original packet capture and evidence securely for further analysis.

11. Limitations

HTTPS application content was encrypted. The investigation does not establish the purpose of all encrypted connections or a relationship between the separate network events. Static imports and strings are not sufficient to classify the executable as malware. No execution or dynamic behavior analysis was performed.

12. Evidence Screenshots

The following screenshots are included as supporting evidence. Captions describe their intended role; the screenshots should be checked visually before submission.

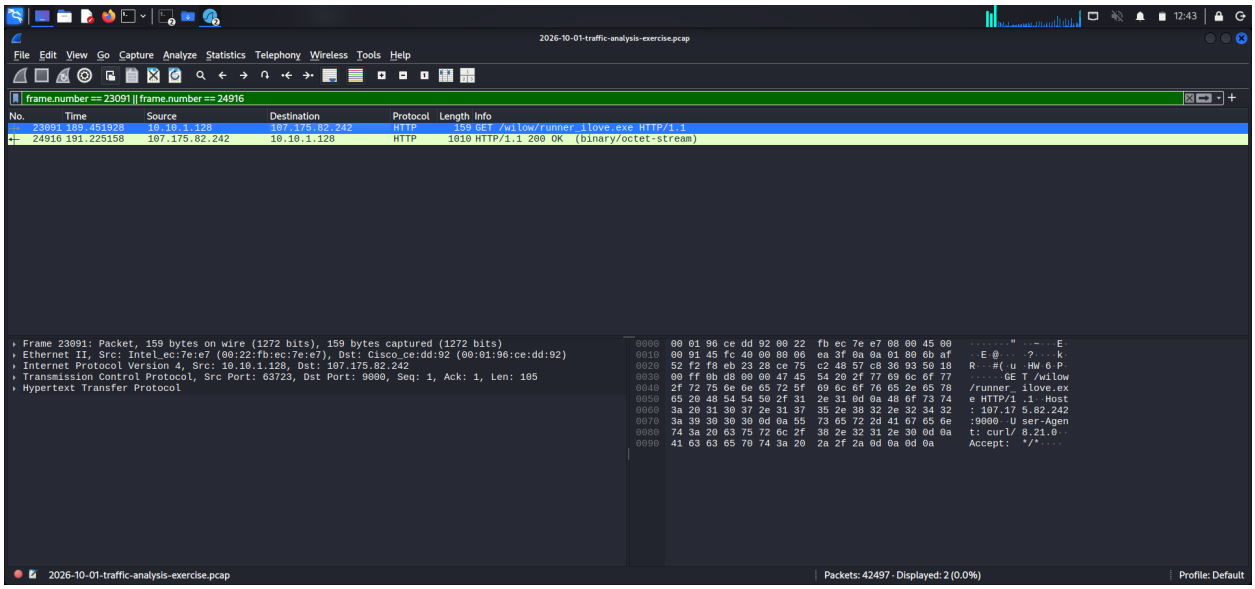


Figure 1. Evidence of the executable download.

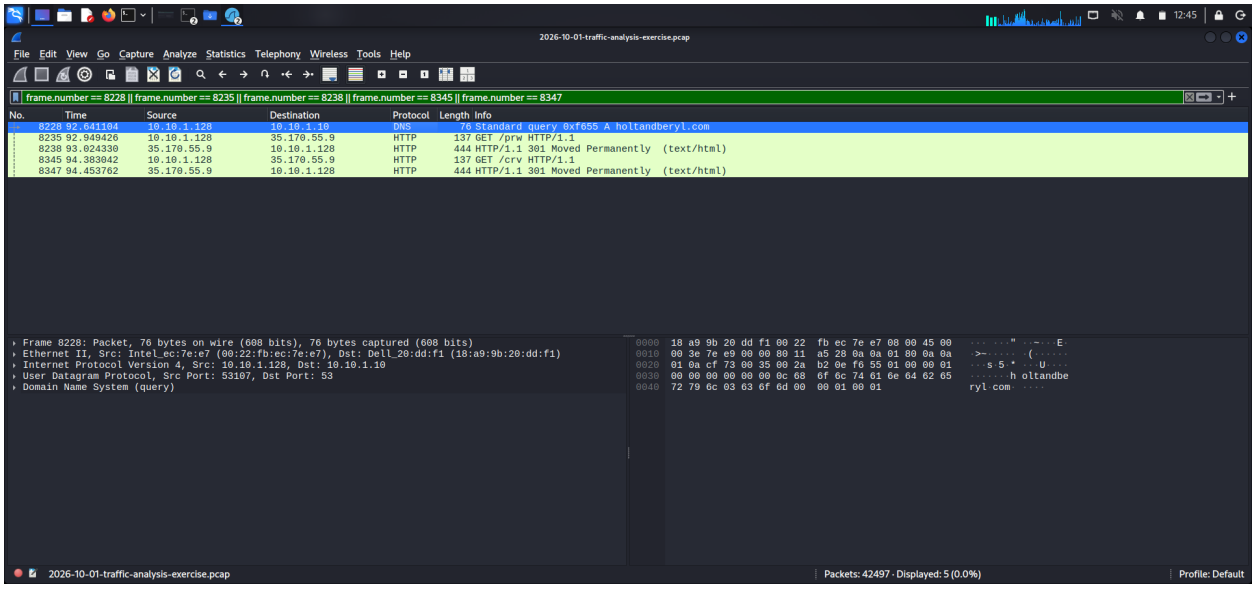


Figure 2. DNS and HTTP investigation evidence.

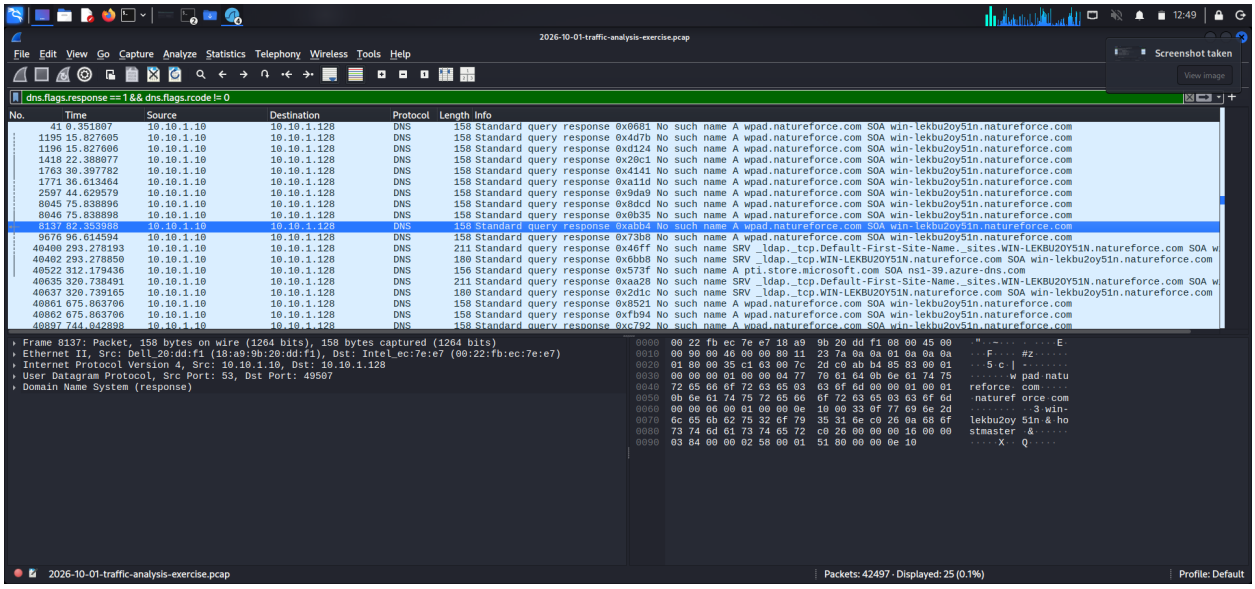


Figure 3. DNS error-response evidence.

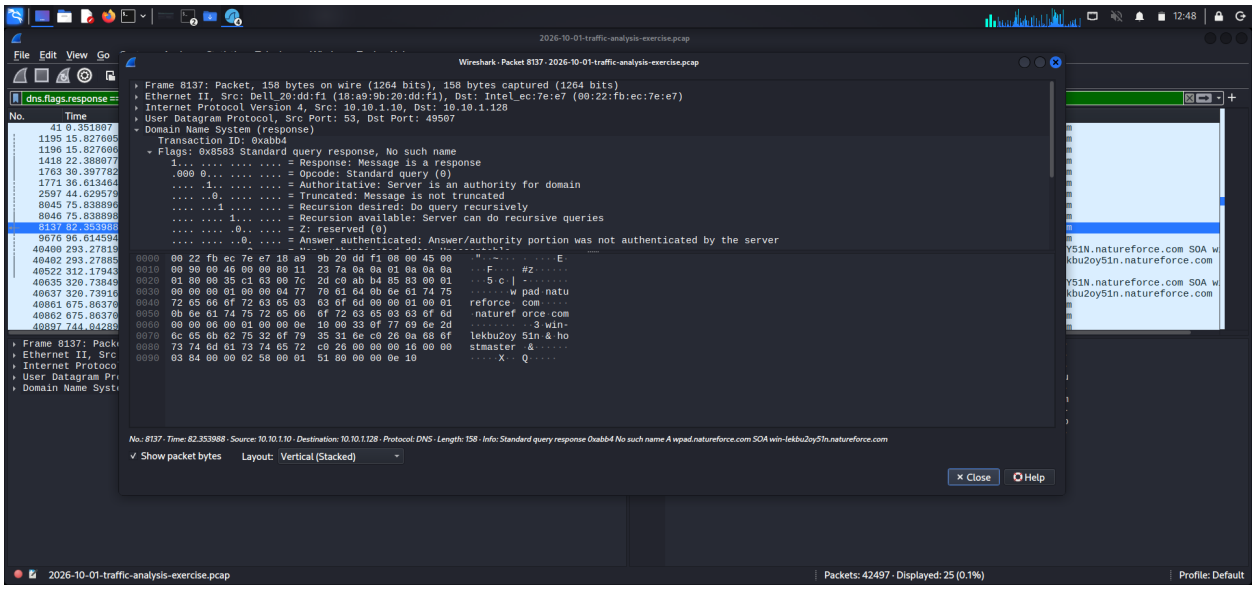


Figure 4. Additional DNS output evidence.

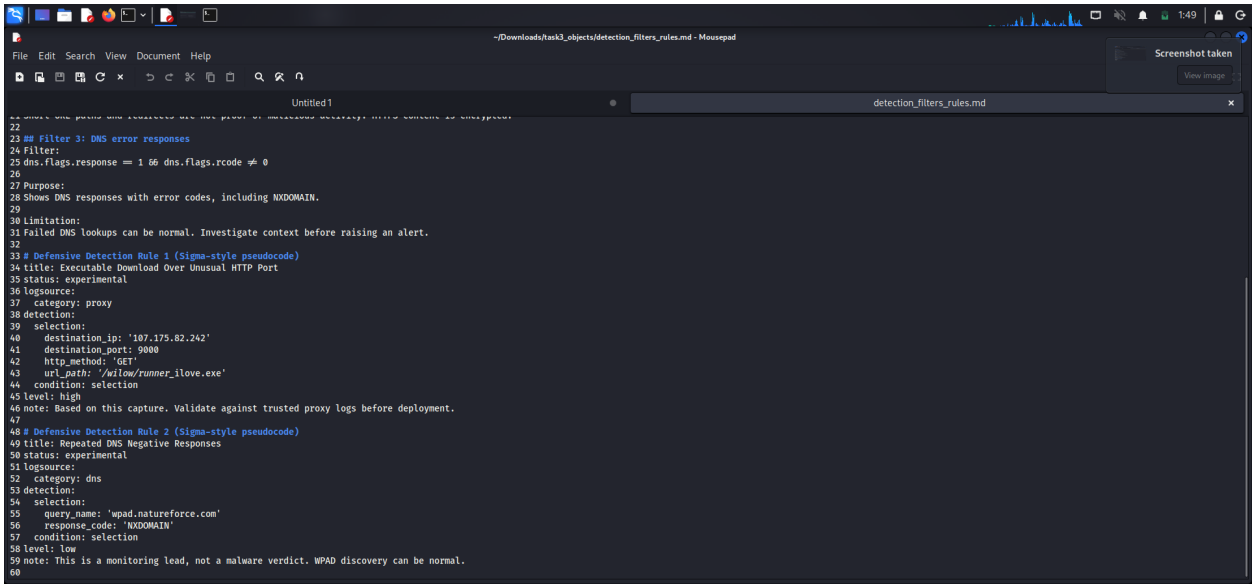


Figure 5. Detection-rule screenshot supplied for the investigation.

13. Supporting Files

The submission folder also contains the IOC table, detection filters and example rules, event timeline, static-analysis notes, and README file.

14. Conclusion

The investigation confirmed an HTTP download of runner_ilove.exe and documented its hash and associated traffic. It also identified additional DNS, HTTP and encrypted connections requiring context. The current evidence does not prove that the executable is malicious or that the observed leads are related. Further endpoint and security-log correlation is recommended before reaching a final verdict.